

PREPARING FOR INFLUENZA SEASON

Data Summary

Objective: *Determine when to send staff, and how many, to each state.*

Hypothesis: *If the size of the vulnerable population (patient is older than 65 years old) increases, then the risk of mortality increases.*

Population Data

Data Sourcing

This is an external data source. The data is provided by the US Census Bureau. As government data, we can assume this as a trustworthy data source.

Data Collection

The data is administrative data estimated as part of the Census Bureau's Population Estimates Program (PEP). It includes estimates of the population across the United States. PEP annually utilizes current data on births, deaths, and migration to calculate population change since the most recent decennial census.

Since the data is an estimate, we should consider that the values are approximate rather than exact representations of the population in the United States.

A potential bias could be the estimation methods used to compile the data.

Data Contents

The dataset contains population counts for the United States from 2009 to 2017. Counts are categorized by county, state, total population, population by gender, and age-specific population grouped into 5-years intervals.

Data Relevance

The data shows the population distribution by age and gender across the United States over multiple years. Since it is estimated by a government agency, it can be considered a trustworthy source of information.

This dataset, combined with the influenza death data can be used to estimate the influenza death rates by age, genre, county and state for the specified years; which can be used to predict future influenza seasons. The inclusion of the age variable is particularly useful for testing the hypothesis that the population aged 65 and older are at

higher risk of dying from influenza. For these reasons, this dataset is essential to achieving the project's goals.

Influenza Laboratory Tests and Patient Visits Data

Data Sourcing

This is an external data source. The data is provided by the Influenza Division at the Center for Diseases Control and Prevention (CDC). As government data, we can assume it is a trustworthy data source.

Data Collection

The data is survey collected, then compiled, and analyzed by the Influenza Division at the CDC, providing year-round information on influenza activity across the United States. The data is voluntary reported in a weekly basis by public health partners and healthcare providers.

Being a voluntary activity, it is important to consider that the dataset may not reflect the actual information but rather an estimated based on the available reports.

Bias may result from various factors, including underreporting, inaccuracy in the reported information, variations in data collection methods and delays in the reporting.

Data Contents

The influenza Visits data tracks patient visits to a medical provider for influenza. It counts the number of visits, number of providers, and total patients seen by week and state from late 2010 to early 2019.

The Lab Test data counts the number of positive influenza laboratory tests by week and state from late 2010 to early 2015.

Data Relevance

Both datasets are relevant to the project, because they provide information about the number of people infected with influenza or seeking medical care, which can help to predict the future influenza seasons. For these reasons, these datasets essential to achieving the project's objective.

Children Flu Shots Data Set

Data Sourcing

This is an external data source. The data is provided by NORC at the University of Chicago under the direction of CDC. As government data, we can assume it is a trustworthy data source.

Data Collection

The data is survey collected through telephone interviews with parents across the United States. The data comes from a random sampling of parents. The demographics are self-reported, but the flu shot information is verified with health providers and can be considered accurate.

Bias may result from various factors, including sampling method, inaccuracies in the reported information, under- or overrepresentation of a population.

Data Contents

The data contains flu shot data from children 6 months old to 17 years old. It is categorized by geographic state and contains family demographics including poverty level, race, and parent marital status.

Data Relevance

This data is relevant for hypotheses concerning children's vulnerability to the influenza virus. However, in this case, it is not relevant for the project's objective or help to testing the hypothesis, therefore, it will not be used.

Sources consulted:

- U.S. Census Bureau. (n.d.). *Population Estimates Program (PEP)*. Retrieved from <https://www.census.gov/programs-surveys/popest.html>
- Centers for Disease Control and Prevention (CDC). (n.d.). *National Influenza Surveillance Overview*. Retrieved from <https://www.cdc.gov/fluview/overview>
- Centers for Disease Control and Prevention (CDC). (n.d.). *About the National Immunization Surveys (NIS)*. Retrieved from <https://www.cdc.gov/nis/about>